Prototype figures from the new export (round 2)

Complex Map Workbench

The Cayley transform $F(z) = (z - i)/(z + i)$ takes the upper half-plane $\mathbb{H}$ onto the unit disk $\mathbb{D}$; it sends i to the center and the point at infinity to 1. Every figure here is placed at its natural size: tick labels are `\footnotesize` and panel labels `\small` of this 10 pt text, set in the same Latin Modern as this paragraph. Lines are vector strokes of at least 0.5 pt; only the shading is an image (flat tones at 600 dpi in indexed color, continuous tone at 300 dpi), and no file declares a font or uses transparency.

Figure 1 is the column preset. Every figure here is placed at its natural size: tick labels are `\footnotesize` and panel labels `\small` of this 10 pt text, set in the same Latin Modern as this paragraph. Lines are vector strokes of at least 0.5 pt; only the shading is an image (flat tones at 600 dpi in indexed color, continuous tone at 300 dpi), and no file declares a font or uses transparency.

Figure 2 is the full-width preset with a detail panel, and Fig. 3 prints the deformation as a filmstrip whose panels can be cited, e.g. Fig. 3(b). Every figure here is placed at its natural size: tick labels are `\footnotesize` and panel labels `\small` of this 10 pt text, set in the same Latin Modern as this paragraph. Lines are vector strokes of at least 0.5 pt; only the shading is an image (flat tones at 600 dpi in indexed color, continuous tone at 300 dpi), and no file declares a font or uses transparency.

Figure 4 is the black-and-white version: 8-bit gray images, 45 % and 14 % gray. Every figure here is placed at its natural size: tick labels are `\footnotesize` and panel labels `\small` of this 10 pt text, set in the same Latin Modern as this paragraph. Lines are vector strokes of at least 0.5 pt; only the shading is an image (flat tones at 600 dpi in indexed color, continuous tone at 300 dpi), and no file declares a font or uses transparency.

Figures 5 and 6 are the Phase view in color and in black and white. Every figure here is placed at its natural size: tick labels are `\footnotesize` and panel labels `\small` of this 10 pt text, set in the same Latin Modern as this paragraph. Lines are vector strokes of at least 0.5 pt; only the shading is an image (flat tones at 600 dpi in indexed color, continuous tone at 300 dpi), and no file declares a font or uses transparency.

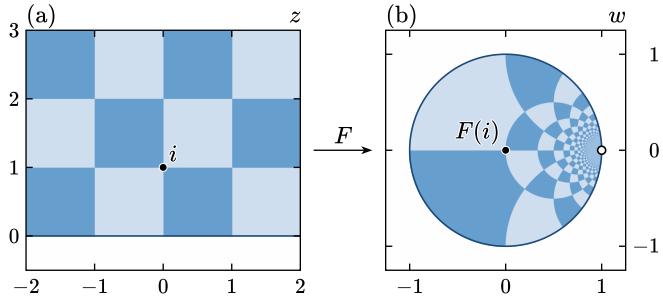


FIG. 1. (a) The upper half-plane $\operatorname{Im} z > 0$ with the point i. (b) Its image under $F(z) = (z - i)/(z + i)$, the unit disk, with $F(i) = 0$. A checkerboard of unit cells in the z-plane is carried by F; the point at infinity goes to 1 (open circle).

Figure 7 uses the overlay. Every figure here is placed at its natural size: tick labels are `\footnotesize` and panel labels `\small` of this 10 pt text, set in the same Latin Modern as this paragraph. Lines are vector strokes of at least 0.5 pt; only the shading is an image (flat tones at 600 dpi in indexed color, continuous tone at 300 dpi), and no file declares a font or uses transparency.

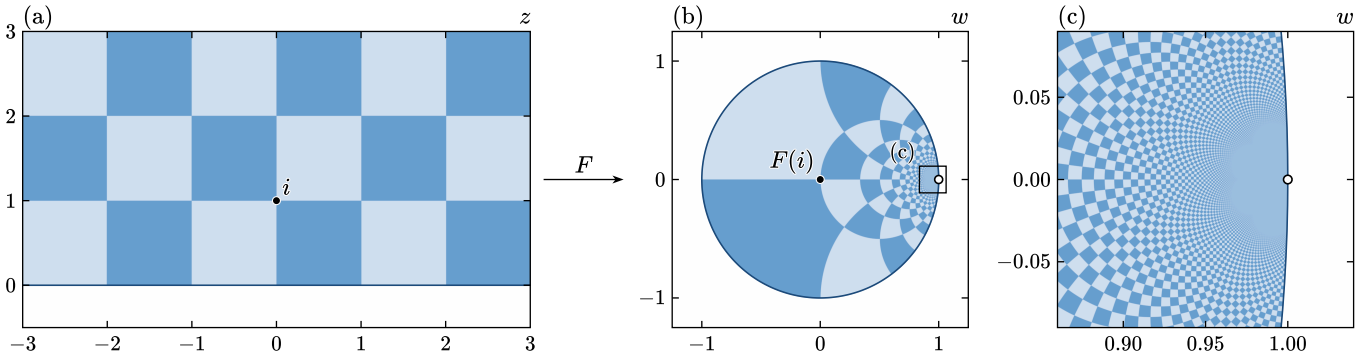


FIG. 2. (a) The upper half-plane $\operatorname{Im} z > 0$ with the point i. (b) Its image under $F(z) = (z - i)/(z + i)$, the unit disk, with $F(i) = 0$; the point at infinity goes to 1 (open circle). (c) Detail of (b) near $w = 1$, where the image cells accumulate. Shading: a checkerboard of unit cells in the z-plane and its image.

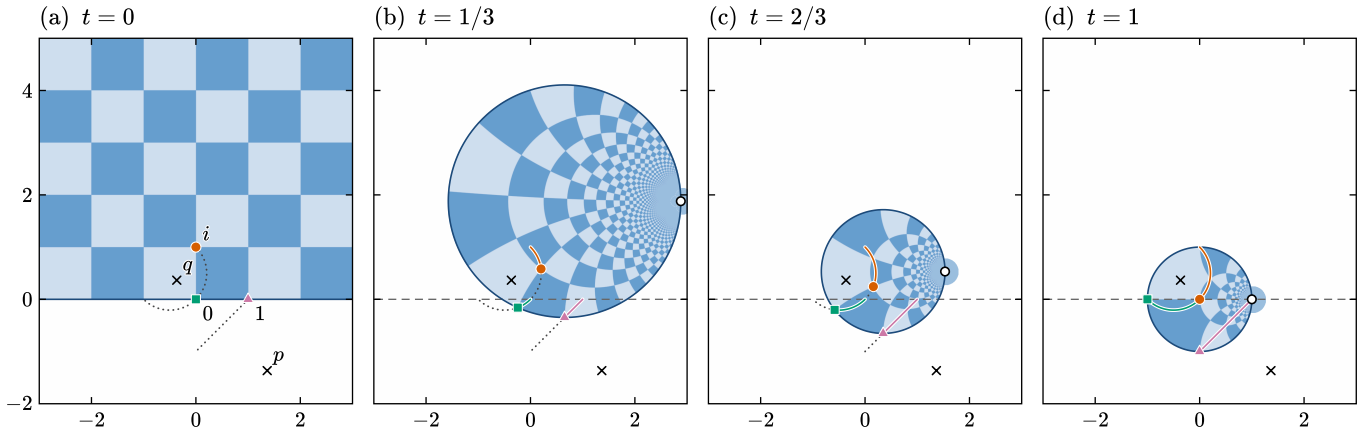


FIG. 3. The Cayley map as a deformation: $H_t(\mathbb{H})$ at (a) $t = 0$, (b) $t = 1/3$, (c) $t = 2/3$ and (d) $t = 1$, where H_t is the elliptic Möbius map fixing $p = \frac{1+\sqrt{3}}{2}(1 - i)$ and $q = \frac{1-\sqrt{3}}{2}(1 - i)$ with $H_t'(p) = e^{2\pi i t/3}$, so $H_0 = \mathrm{id}$ and $H_1 = F$. Solid: the paths $t \mapsto H_t(z)$ of $z = i$ (circle), 0 (square) and 1 (triangle) up to the frame's t; dotted: the rest of each path; dashed: the real axis; open circle: $H_t(\infty)$.

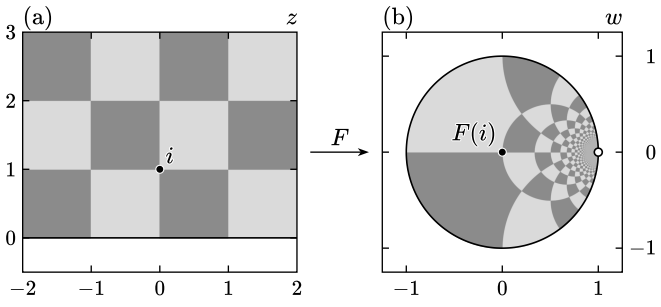


FIG. 4. (a) The upper half-plane $\operatorname{Im} z > 0$ with the point i. (b) Its image under $F(z) = (z - i)/(z + i)$, the unit disk, with $F(i) = 0$. A checkerboard of unit cells in the z-plane is carried by F; the point at infinity goes to 1 (open circle).

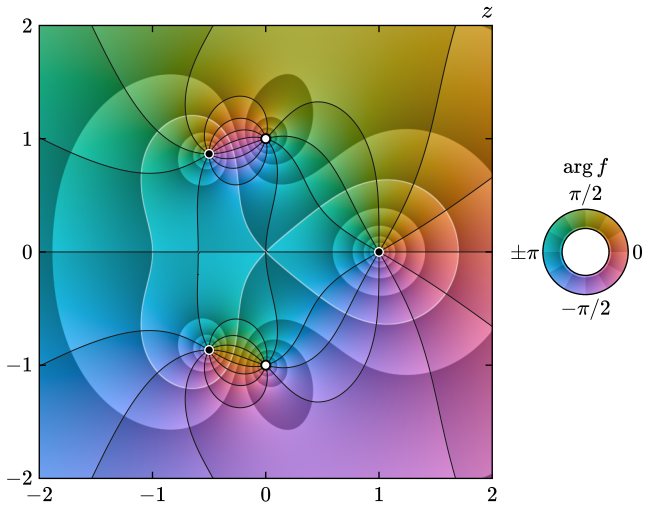


FIG. 5. Phase portrait of $f(z) = (z^3 - 1)/(z^2 + 1)$ on $[-2, 2]^2$: the color shows $\arg f(z)$ (key at right) and the thin lines are the level sets $\arg f \in \frac{\pi}{6}\mathbb{Z}$. Zeros (dots) at the cube roots of unity; simple poles (open circles) at $\pm i$.

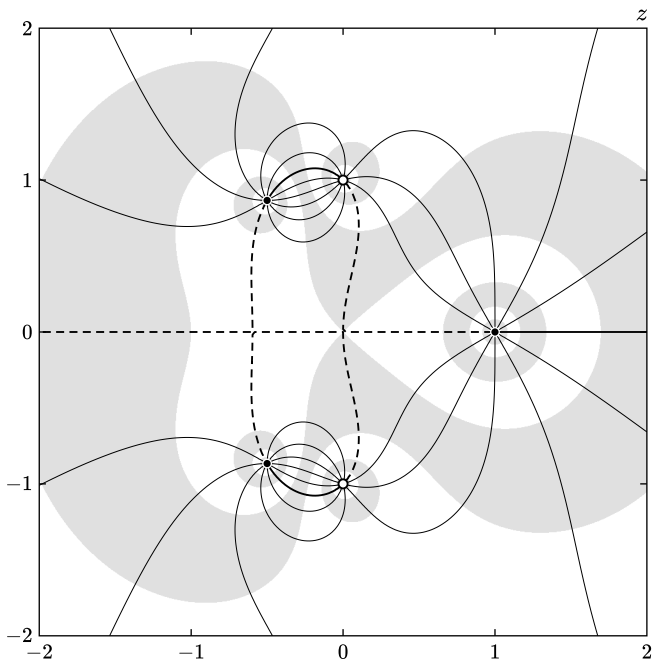


FIG. 6. Phase portrait of $f(z) = (z^3 - 1)/(z^2 + 1)$ on $[-2, 2]^2$ as level sets of $\arg f$ at multiples of $\pi/6$; heavy: $\arg f = 0$, dashed: $\arg f = \pi$. Bands: $\lfloor \log_2 |f| \rfloor$ even (gray) or odd (white). Zeros (dots) at the cube roots of unity; simple poles (open circles) at $\pm i$.

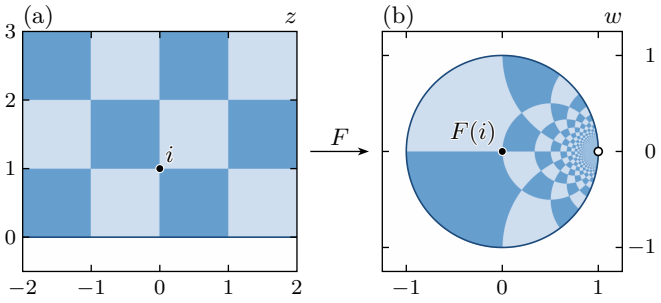


FIG. 7. The overlay format: the art of Fig. 1 with every label typeset by this document. (a) The upper half-plane $\operatorname{Im} z > 0$ with the point i. (b) Its image under $F(z) = (z - i)/(z + i)$, the unit disk, with $F(i) = 0$. A checkerboard of unit cells in the z-plane is carried by F; the point at infinity goes to 1 (open circle).